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Do security breaches matter? The shareholder puzzle

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Abstract

This article analyzes the effect of computer breaches on publicly traded equities from 2005 to 2017. An event study is performed and breaches analyzed conditioned on whether the breach announcement has been made in the mainstream media or through other channels. We find that in the period prior to the announcement date in the media, the mean abnormal return is negative, reflecting a likely leakage of information. In the period following the announcement date, the mean abnormal return is positive, often more than offsetting the previous declines. The findings have important implications for analysts, portfolio managers, institutional investors, and regulators.

KEYWORDS

computer breaches, event study, information leakage, regulation inconsistencies

JEL CLASSIFICATION

*G*14, *G*18

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1 | INTRODUCTION

Computer breaches have received significant attention in the financial press, yet little attention in the academic financial literature. Investors and analysts currently have little guidance as to the expected effects of breaches on security prices. To understand the effect of breaches on publicly traded equity issues, we analyze all such available breach data by type and by industry from 2005 to 2017. We find that typically in the period immediately before the announcement of a breach in the media, the mean cumulative abnormal return (CAR) is negative, while immediately following the announcement, the mean CAR is positive, often more than offsetting the prior decline.

Currently, when analysts, portfolio managers, and institutional investors hear about a security breach, they have very little guidance as to the potential impact of the breach on the breached company's stock. Surprisingly, it will be shown that in general, following the breach announcement, one is more likely to observe an increase rather than a decrease in the stock price. This is counterintuitive to the expectations of analysts, portfolio managers, and investors. Moreover, a negative CAR in the period prior to the breach announcement should serve as a flag to regulators. The most likely reason for a decline is information leakage. The Securities and Exchange Commission (SEC) is regularly analyzing any abnormal trading activity prior to announcements of mergers and acquisitions and, for decades, it has been well known that insider trading activity may take place prior to these M&A announcements. This article suggests that the regulators should also consider a routine trading history analysis of the stock of companies that have been breached. The focus of regulators' attention should be the several weeks prior to the breach announcement.

2 | LITERATURE REVIEW

Much of the literature discussing data breaches focuses on the informational aspects of the breach, with the vast majority of the papers published in the management information system literature. Several are focused on the costs and determination of methodologies to remedy the occurrence of breaches. For example, Gwebu, Wang, and Xie (2014) determine that the damage caused by a data breach primarily stems from direct costs such as compensation or litigation costs rather than diminished market share or sales. Goode, Hoehle, Venkatesh, and Brown (2017) assess the intention to repurchase product following data breaches at the Sony PlayStation Network.

Several papers published in information security journals also assess the results of event studies surrounding breaches. Pirounias, Mermigas, and Patsakis (2014) find that firms incur negative statistically significant returns resulting from breaches and that technology firms tend to incur higher costs in breaches than non-technology firms. Schatz and Bashroush (2016) find that no strong conclusions can be drawn relating the impact on the market value of multiple data breaches. In an early work, Cavusoglu, Mishra, and Raghunathan (2004) find that the breached firm lost an average 2.1% of market value within 2 days of the announcement of the breach. Campbell, Gordon, Loeb, and Zhou (2003) in another early paper find that there is a significant negative reaction to a breach when there is a loss of confidential information and no significant market reaction when the breach does not involve confidential information. While the prior papers find that there is generally a significant negative reaction to breaches, Kannan, Rees, and Sridhar (2007) find no significant abnormal return around the breach date. Hovav

and Gray (2014), using a stakeholder analysis, find that while some stakeholders lose, others win as a result of a breach. Their analysis suggests that the effect of the breach on the attacked firm varies over time. Arcuri, Brogi, and Gandolfi (2017) find that substantial negative returns follow the announcement of breaches, with financial firms incurring the greatest negative return. They also posit a link between cybercrime and insider trading. Bose and Leung (2014) study phishing in 42 countries and conclude that a significant loss was incurred around the announcement date of the breach. Chen, Dong, Chen, and Xu (2016) determined that there is no difference in the effect on market values of a breach announcement for firms cross-listed and those not cross-listed. Amir, Levi, and Livne (2018) find that firms underreport information on cyberattacks.

Although breaches are not uniquely defined in the finance literature, several topics associated with breaches are discussed in management journals. Choong, Hutton, Richardson, and Rinaldo (2017) find that both the direct and indirect costs of a breach are difficult to assess. They use an event study to determine that a small but significant negative abnormal return is associated with the announcement of a breach. Kvochko and Pant (2015) find that even major breaches had only a slightly negative impact on companies' stock prices and that following the breaches, there often were quick recoveries.

3 | TYPES OF BREACHES

While breaches cover a broad range of system incursions, we subdivide them into four types: traditional hacks, negligence, phishing, and theft.

The traditional hack is defined as the use of a computer to gain unauthorized access to a system. Using this definition, we were able to identify a broad range of hacks (see, e.g., Clements, 2018; Germano, 2018; Isaac, Benner, & Frenkel, 2017; Meyer, 2016; White, 2017). For example, these include a hack of Twitter in 2016, resulting in 32 million passwords stolen from accounts and sold on the dark web. Another hack involved improperly accessing data from 150 million users of Under Armour's MyFitnessPal platform in early 2018. In this case, emails, usernames, and passwords were exposed. This was also the type of breach used at Equifax to obtain highly personal data such as social security numbers, addresses, and birth dates of 147.9 million US consumers in a hack that began in May 2017. The Equifax hack was identified several months later, on 29 July 2017, and then made public 5 weeks later, on 7 September. One can comprehend the magnitude of the Equifax hack by realizing that the 147.9 million people hacked are more than the approximately 125 million households in the United States. In another hack, the drivers' licenses of 600,000 Uber drivers and approximately six dozen users of the Uber app were stolen in 2016. Interestingly, Uber did not reveal the hack until approximately 1 year later and in the interim paid the hackers to destroy the data, yet they made the payment with no evidence that the data was destroyed. Recently, in July 2019 in one of the largest-ever thefts of bank data, a hacker compromised the personal information (e.g., names and addresses) of over 100 million Capital One customers.

A breach due to negligence occurs when there is a leak of personal information due to the lack of company oversight. It could be anything ranging from improper disposal of client personal information to accidental dissemination of such information (e.g., wrongly emailing client data) or missing computer hardware containing client personal information. For example, an AT&T breach in 2014 exposed approximately 280,000 customer names with full or partial social security numbers. The breaches occurred at overseas call centers when employees

without appropriate access privileges obtained customer social security numbers so they could sell the information to thieves requiring the numbers to unlock stolen cell phones. AT&T agreed to pay a $25 million penalty to the Federal Trade Commission stemming from its negligence (see, e.g., Boxall, 2015).

Phishing occurs when an attacker obtains sensitive information (e.g., usernames, passwords, or credit card details, etc.) for malicious means often using a skimmer or similar device to collect credit card details belonging to unsuspecting customers. In addition, there are also breaches whereby the attacker impersonates a customer, usually for monetary gain (e.g., selling information on the black market, making unauthorized withdrawals, etc.). In a common phishing breach, an email is sent to unsuspecting customers of major retailers such as Walmart or Amazon, telling the customer that she has a short period of time to fill in the requested information if she wants to have the option of returning a product without a fee. A classic early phish request was an email that was sent during the 2013 Christmas season emulating an email from Walmart. The email indicated that if the requisite personal information was not filled out within a week, a 17% fee would be deducted from a product's price should the customer wish to return it (see, e.g. 'Be aware of phishing scams involving Walmart emails', December 23, 2013). Such phishing appears realistic and provides an incentive for the unsuspecting customer to quickly return the requested information. More recently, in April 2018, Saks Fifth Avenue and Lord & Taylor revealed that they had been stung by a phishing attack between May 2017 and March 2018. The perpetrators installed malicious software on the cash registers, enabling the criminals to obtain more than 5 million card numbers from all cards swiped at that machine. The software was installed by employees inadvertently clicking on attachments sent by those involved in setting up the phishing breach. The group involved in the theft has been identified as the Russian speaking Fin7 or JokerStash and following the breach have offered 125,000 of the records for sale (see, e.g., McMillan & Kapner, 2018).

A breach due to theft occurs when a document, computer, or computer file/hardware containing sensitive information is stolen from a company. Most of the breaches involve the theft of computer hardware, with several cases involving employees stealing customer files and selling the information on the black market or to outside vendors. In January 2015, for example, Morgan Stanley revealed that an employee had stolen account information belonging to up to 10% of its wealth management clients. It was reported in Forbes that the bank discovered the customer data on the file-sharing site Pastebin and that the employee was planning to sell the client data (see, e.g., Din, 2015).

4 | BREACH DISCLOSURE REGULATION INCONSISTENCIES

It is interesting to note that there is no common federal requirement to publicly reveal which companies have been breached or, if breached, how and when to notify those whose records were breached. In other words, each state has its own rules as to whether, when, and how to notify victims whose records were breached (see, e.g., Security Breach Notification Chart, 2017). While some notification rules are identical, most are different. For example, California, New York, and Georgia have the most breaches affecting publicly traded equities during our investigation period, respectively, 62, 44, and 24 breaches, and their notification laws are identical. Namely, the notification laws for these states are for all firms breached that are doing business in the state or even if the breached company simply maintains information on state

residents. These laws differ from Illinois, the next most heavily breached state (23 breaches), whose breach statute indicates that its breach notification law affects breaches to 'any entity that, for any purpose, handles, collects, disseminates, or otherwise deals with non-public personal information even if it does not conduct business in the state.' The notification law for the state of Washington breaches (17 breaches) affects all companies doing business in the state. Thus, while the notification laws in different states are similar, they are different in significant respects. While a number of states indicate that the notification to those breached or to the state should be made in as 'expedient time as possible without unreasonable delay' (California, New York, Georgia), Illinois is more specific indicating that notification should be in the most expedient time possible, without unreasonable delay, but adding that within 5 days of learning of the breach, it must inform the state general assembly, and within 45 days inform the state attorney general if it affected more than 250 state residents. Florida indicates that companies should be as expeditious as possible in communicating with the victims, and communicate with them in less than 30 days, while Connecticut indicates that it should be communicated without unreasonable delay, but not later than 90 days after learning of the breach. Such variability in state notification regulations results in significant autonomy for each company as to when it reveals the breach to either those affected or the state.

The actual mechanics of the process by which notification to those individuals whose records were breached varies by state as well. For example, California requires that every state resident whose data was stolen be informed, and if the breach affected more than 500 state residents, the state attorney general must be notified. In New York, all those whose data was breached must be informed and state attorney general notification is mandatory, while consumer reporting agencies must be notified if the breach affected more than 5,000 state residents. In both Georgia and Texas, while state residents must be informed, there is no attorney general notification mandated, yet consumer reporting agencies must be notified if more than 10,000 records were compromised. Each state also has an individual substitute notification plan if notification becomes too costly or impossible.

5 | DATA

The data for this study was obtained from the Privacy Rights Clearinghouse (PRC), a non-profit consumer education and advocacy organization. PRC aggregates data from a number of sources including Dataloss DB, state attorney generals, government agencies, Databreaches.net, and a range of other providers. Based on the PRC data, we were able to identify 5,331 instances of breaches between 1 January 2005 and 23 February 2017. While most of these breaches occurred at private or non-profit organizations, the purpose of this study is to analyze the effect of breaches on the equity value of public companies. Thus, the PRC breach data was disaggregated into segments relevant for this analysis. Moreover, the PRC data is classified into types of breaches, including hacking, insider breach, unintended disclosure, physical loss, portable device breach, and stationary device breach. While the growth of several of these types has been dramatic (e.g., hacking), others have diminished markedly over time (e.g., stationary device and physical loss) and were not amenable to statistical analysis (see, e.g., Williamson, 2015). As a result, we reviewed each breach classification and reclassified the breaches of public companies into a revised set of four categories: hacking, negligence, phishing, and theft. All public breaches in our dataset were assigned to these four breach categories.

TABLE 1 Sample selection waterfall

This table reports the sample selection process. The first row indicates the total number of data breaches. Then each successive row indicates a particular reason those breaches could not be incorporated in the analysis. For example, row 2 identifies the major reason breaches were eliminated from the dataset, namely the breaches occurred at a non-public company. The bottom row indicates the size of the final sample.

Selection criteria	# Breaches
Initial number of data breaches from PRC database	5,331
Removing breaches at non-public companies	(4,939)
Removing companies not public at time of breach or if data was present in Bloomberg	(45)
Removing duplicate breaches	(3)
Final sample	344

There were a total of 344 breaches of public companies during the investigation period. The breaches were spread relatively evenly over the 12-year period, with 164 breaches in the first 6 years, from 2005 through 2010 and 177 breaches in the second 6-year period, from 2011 through 2016, with 3 in the 2-month stub period, 2017.

To obtain our database of 344 public company breaches, we began by enumerating all the breaches in the investigation period (5,331 breaches) and eliminating breaches for which no public company data was available. The data inclusion waterfall can be observed in Table 1.

Interestingly, most of the breaches occurred at private institutions such as hospitals, governmental bodies, and privately held companies (4,939 breaches). In addition, data for 45 breaches was unavailable on Bloomberg. There were also three breaches which were double counted in the database and were eliminated.

Moreover, to assess whether the publication of the news of the breach in major media outlets is associated with a market value reaction, we subdivided the data into whether the breach was

TABLE 2 Sample distribution by industry

This table reports the distribution of firms in the sample by industry. The first column indicates the industries included in the analysis. The second column indicates the number of breaches in each industry reported in the mainstream media (e.g., *Business Week*, *The Wall Street Journal*, *New York Times*, etc.). The third column indicates the number of breaches not announced in the mainstream media. The fourth column represents the sum of the second and third columns and indicates the total number of breaches included in the sample by industry.

Industry	In media	Not in media	Total
Banks	16	46	**62**
Large chain stores	17	56	**73**
Miscellaneous	17	61	**78**
Other financial institutions	12	56	**68**
Technology	17	22	**39**
Telecommunications	6	18	**24**
Total	**85**	**259**	**344**

reported in the mainstream media (e.g., *Business Week*, *New York Times*, *The Wall Street Journal*, etc.) or whether it was simply reported through other sources. To determine if a breach was reported in the mainstream media, we reviewed Factiva for each of the 344 breaches. Of these 344 breaches, 85 were reported in the media and 259 were not.

The breach data was analyzed by both industry and breach type and was also classified by whether the breach was reported in the media or not. The number of breaches by industry and breach type are presented in Tables 2 and 3, respectively.

To understand the order of magnitude of the breaches, we assessed the number of records breached. While in most cases, the number of records was not revealed, there appears to be a skewed distribution of the number of records breached. While there were 104 breaches in which there were fewer than 10 K records compromised during the investigation period, there were 18 breaches in which over 1 million records were compromised and an additional 18 breaches in which between 100,000 and 1 million records were affected.

6 | METHODOLOGY

In order to assess the extent of breaches on publicly held equities, a wide variety of statistical tests were employed. Using the event study methodology, abnormal returns (ARs) and cumulative abnormal returns (CARs) were derived for a period of 10 days before and 20 days after the announcement date. For calculations of stock market abnormal returns, see, for example, Bulkey and Herrerias (2005), Carretta, Farina, and Schwizer (2010), and Michel, Oded, and Shaked (2014).

Unlike typical event studies applied to mergers, stock splits, or dividend announcements, in the case of a breach, defining the announcement date requires additional analysis. In the case where the breach is announced in the mainstream media ('in media'), the announcement date is simply the date of the announcement in the media. These media announcement dates were all confirmed using the Factiva database. For breaches that have not been announced in the media ('not in media'), the announcement date has been reported in the PRC. This is the date determined by the PRC to be when the breach was made public. For example, if the source of the breach information was the Maryland Attorney General, it would be the date at which the

TABLE 3 Sample distribution by breach type

This table reports the distribution of firms in the sample by breach type. The first column lists all the types of breaches analyzed. The second column indicates the number of a specific breach type (e.g., negligence) announced in the mainstream media. The third column indicates the number of a specific breach type not announced in the mainstream media. The fourth column is the sum of the second and third columns and represents the total for each breach type in the sample.

Breach type	Media	Not in media	Total
Negligence	23	103	**126**
Theft	15	82	**97**
Hack	40	47	**87**
Phishing	7	27	**34**
Total	**85**	**259**	**344**

information was received by the attorney general's office. Unfortunately, as was described above in section 4, the timing of breach reporting is not standardized across states. Thus, for in media cases, the announcement in certain states can be immediately following the company learning of the breach, yet in other cases, it can be months following the company being apprised of such a breach. In other words, it is possible that there is a time delay between when the company learns of the breach and the time it reports the breach to the state and those affected.

Within the 30-day time interval surrounding the announcement (10 days before and 20 days following the announcement), we considered 10 time windows. These windows included standard 3- and 5-day event windows in proximity of the announcement date (e.g., (−1, 1)) as well as others, and also included extended windows beginning immediately following the announcement date through the end of the observed time interval (e.g., (1, 20)) as well as through the end of an intermediate time interval (e.g., (1, 10)). The motivation for the extended time windows is to allow for the possibility that the time required for the market to assess the impact of a breach might be longer than the market assessment of standard corporate events (e.g., stock splits, mergers, and dividends). For robustness, in performing our analysis, we derived both Capital Asset Pricing Model (CAPM)-based CARs and four factor-based CARs (henceforth FF-based CARs). The four factor model includes the Fama–French three factor model plus the Carhart momentum factor.

We applied statistical tests based on assumed normality, and also performed a distribution-free non-parametric test. Specifically, the parametric tests run are the Patell test (also referred to as the standardized residual test), the standardized cross-sectional test (also referred as the Boehmer, Musumeci, and Paulsen test), and the skewness corrected test; while the non-parametric test is the sign test. For a more thorough discussion of the statistical analysis used in event studies, see Schimmer, Levchenko, and Müller (2015) and Rani, Yadav, and Jain (2016).

The data was analyzed in six different industry categories. These include all breaches, breaches in banks, large chain stores, technology firms, telecommunications firms, and other financial institutions. Also, as described earlier, we analyzed the following four types of breaches: hacking, phishing, theft, and negligence. In addition, for each analysis, there were two categorizations: those breaches announced in the media and those not announced in the media. To obtain some degree of statistical significance we only analyzed categories with more than 15 occurrences.

7 | RESULTS

There are two primary observations that are evident in the results. The first is that around the announcement date of the breach in the media, there is a tendency of the market to anticipate the breach. There is often a statistically significant dip just prior to the announcement date and over an event window tightly constructed around the announcement date. While this does not occur in all categories, it is frequently observed. The second observation is that from immediately after the announcement in the media, over the next 10–20 days, the CARs often reflect a statistically significant increase in value. The results are qualitatively identical for both the CAPM-based and the FF-based CARs for virtually all categories. These results are much less pronounced and, in some cases, nonexistent for breaches not announced in the media.

To better understand the nature of the analysis performed, as stated before, we present the statistical and graphical results for the following categories in detail: all breaches (combining

TABLE 4 Abnormal returns for all all breaches based on parametric and non-parametric tests

Panel A: This table reports the cumulative abnormal returns determined using the CAPM and includes all breaches in the sample, namely both those announced in the media and those not announced in the media. The event windows include those structured around the announcement date and those in the post-announcement period. The first column indicates the event windows investigated. Four statistical tests were run to assess the significance of the CAPM-based CARs. The second column includes results of the Patell test, the third column describes the results of the standardized cross-sectional test, the fourth column describes the results of the skewness corrected test, and the fifth column includes results of the non-parametric sign test. ** and * denote significance levels of 1% and 5%, respectively.

Event window	Patell test	Standardized cross-sectional test	Skewness corrected test	Sign test
(−3, 1)	−2.01*	−1.85	−1.57	−1.29
(−2, 1)	−2.96**	−2.54*	−2.28*	−2.48*
(−2, 0)	−2.28*	−2.19*	−2.02*	−1.40
(−2, −1)	−2.12*	−2.15*	−1.89	−1.73
(−1, 1)	−2.85**	−2.38*	−1.97*	−1.19
(0)	−0.96	−0.95	−1.03	−1.19
(1, 10)	0.65	0.67	0.06	0.75
(1, 20)	1.41	1.44	1.25	0.97
(2, 10)	1.35	1.42	0.53	1.19
(2, 20)	1.90	1.94	1.65	1.29

Panel B: This table reports the cumulative abnormal returns determined using four factors and includes all breaches in the sample, namely both those announced in the media and those not announced in the media. The event windows include those structured around the announcement date and those in the post-announcement period. The first column indicates the event windows investigated. Four statistical tests were run to assess the significance of the CAPM-based CARs. The second column includes results of the Patell test, the third column describes the results of the standardized cross-sectional test, the fourth column describes the results of the skewness corrected test, and the fifth column includes results of the non-parametric sign test. ** and * denote significance levels of 1% and 5%, respectively.

Event window	Patell test	Standardized cross-sectional test	Skewness corrected test	Sign test
(−3, 1)	−1.65	−1.46	−1.14	−0.97
(−2, 1)	−2.60**	−2.13*	−1.82	−1.51
(−2, 0)	−2.10*	−1.91	−1.74	−1.40
(−2, −1)	−1.86	−1.77	−1.47	−1.40
(−1, 1)	−2.57*	−2.05*	−1.66	−1.51
(0)	−1.00	−0.95	−1.10	−0.54
(1, 10)	0.81	0.80	0.54	1.94
(1, 20)	1.40	1.37	1.38	0.97
(2, 10)	0.00	1.38	0.88	2.37*
(2, 20)	1.80	1.75	1.64	1.08

breaches announced in the media and those not announced in the media), and all breaches, breaches in banks, large chain stores, technology firms, and telecommunication firms (each separately categorizing breaches announced in the media and not announced in the media), and hacking, phishing, theft, and negligence (each separately analyzing those announced in the media and those not announced in the media). For several other breach categories, because of similarity to other results, we will discuss the results without graphical or statistical display. Because the results using CAPM-based CARs and FF-based CARs are qualitatively identical, we report the results using both methods only for the complete sample. For the sake of brevity in the rest of the analysis, we report only the CAPM-based analysis in all tables following Table 4.

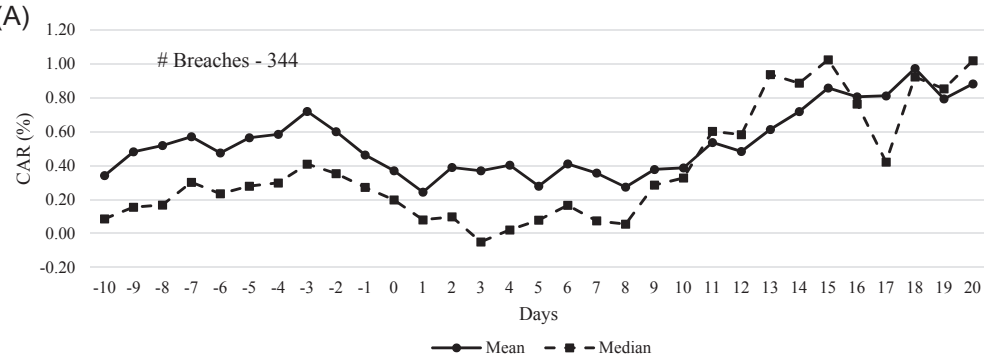
7.1 | All breaches

Table 4 reports our finding for the complete sample (all breaches) using the three parametric tests and the non-parametric test. The results are reported for the CAPM-based CARs and the FF-based CARs in Panels A and B, respectively. Using the CAPM (Table 4, Panel A), the statistical results pertaining to each of the event windows in the proximity of the announcement date, namely (−1, 1), (−3, 1), (−2, 1), (−2, 0), (−2, −1), suggest that there is a statistically significant drop in value in anticipation of the breach announcement. These results differ from those described by Arcuri et al. (2017), who find that substantial negative returns follow the announcement of breaches. Our observations are based on the results of the Patell test, the standardized cross-sectional test, and the skewness corrected test. Most of these drops are statistically significant at the 5% level or in some cases at the 1% level. In addition, several of the drops are also significant using the non-parametric sign test (significant at the 5% level for the (−2, 1) interval and 10% for the (−2, −1) interval). While most of the windows in the proximity of the announcement date are statistically significant, it is noteworthy to observe that the window consisting of pure leakage (−2, −1) is also statistically significant and negative. In other words, in the 2-day window prior to the actual announcement, there is a statistically significant drop in the mean CARs. To assess whether this result is driven by media announcements, in the following subsection we test whether there is a difference in the mean CARs for this pure leakage window when announcements are made through the media and when they are announced through other outlets (e.g., through the state where the breach occurred). (Throughout the paper, while we refer to results which are statistically significant at the 10% level, the tables indicate (via asterisks) only the ones that are of significance at the 5% and 1% level.)

Interestingly, immediately following the announcement, the cumulative abnormal returns begin to increase in value and over approximately the next 20-day period (2, 20), the mean CARs become marginally statistically significant (at the 10% level). In other words, following a drop in the value of mean CARs in anticipation of the breach announcement, the cumulative abnormal returns regain their lost value and become positive. These results are supported by Kvochko and Pant (2015) who find that following the breaches there often were quick recoveries. For robustness in Panel B we repeat the analysis in Panel A using FF-based CARs instead of CAPM-based CARs. As can be observed, the results using the FF-based CARs are qualitatively identical to the results using the CAPM-based CARs.

The cumulative abnormal returns can also be examined using the graphs of CAPM- and FF-based CARs which are presented in Figure 1, Panels A and B, respectively. In Figure 1, Panel A, the CAPM results indicate that the mean drop in value from day −2 through day 1 is −0.48% (median drop is −0.33%). While a seemingly small percentage, this drop as well as other drops

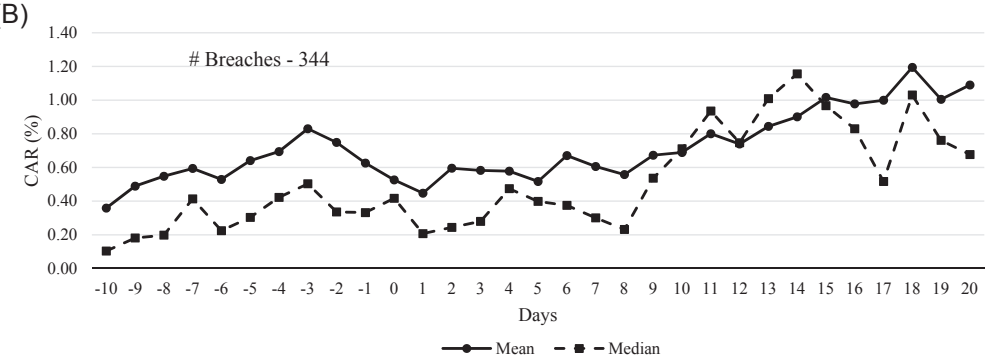
(A)



Day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mean	0.34	0.48	0.52	0.57	0.47	0.56	0.58	0.72	0.60	0.46	0.37	0.24	0.39	0.37	0.40	0.28	0.41	0.36	0.27	0.38	0.39	0.54	0.48	0.61	0.72	0.86	0.81	0.81	0.97	0.79	0.88
Median	0.09	0.16	0.17	0.30	0.23	0.28	0.30	0.41	0.35	0.27	0.20	0.08	0.10	-0.05	0.02	0.08	0.17	0.07	0.06	0.29	0.33	0.60	0.58	0.94	0.89	1.02	0.76	0.42	0.92	0.85	1.02

This figure describes the evolution of both the mean and median cumulative abnormal returns calculated using the CAPM from 10 days prior to the announcement of the breach through 20 days following the breach. It includes all of the breaches in the sample, including both those announced in the media and those not announced in the media. The total number of breaches described in the figure is 344.

(B)



Day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mean	0.36	0.49	0.55	0.59	0.53	0.64	0.69	0.83	0.75	0.63	0.53	0.45	0.60	0.58	0.58	0.52	0.67	0.61	0.56	0.67	0.69	0.80	0.74	0.84	0.90	1.02	0.98	1.00	1.19	1.00	1.09
Median	0.10	0.18	0.20	0.41	0.22	0.30	0.42	0.50	0.34	0.33	0.42	0.21	0.24	0.28	0.47	0.40	0.37	0.30	0.23	0.54	0.71	0.93	0.75	1.01	1.16	0.97	0.83	0.52	1.03	0.76	0.68

This figure describes the evolution of both the mean and median cumulative abnormal returns calculated using four factors from 10 days prior to the announcement of the breach through 20 days following the breach. It includes all of the breaches in the sample, including both those announced in the media and those not announced in the media. The total number of breaches described in the figure is 344.

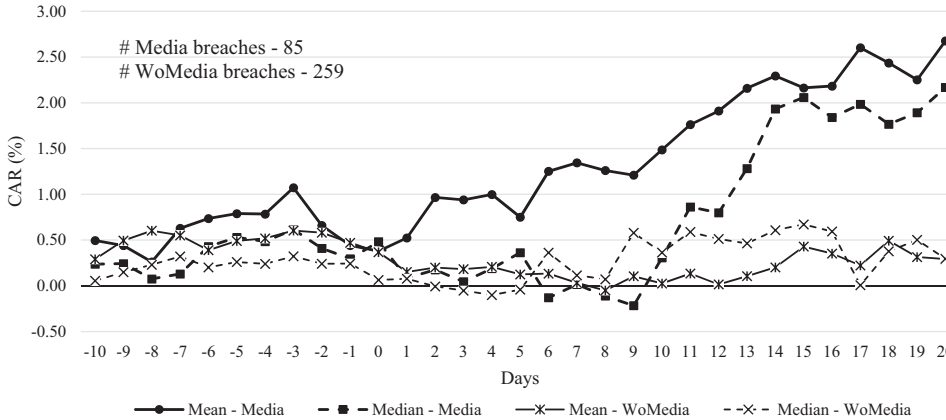
FIGURE 1 Panel A: All breaches (in media and not in media combined) – CAPM CARs. Panel B: All breaches (in media and not in media combined) – four factor CARs

in the period surrounding the announcement date (e.g., (−1, 1), (−2, 0)) are seen to be statistically significant using a number of the tests in the statistics panel presented in Table 4, Panel A (see, e.g., Patell test, standardized cross-sectional test, and the skewness corrected test). Moreover, this translates to a sizeable mean dollar drop of −$348.2 million and a median dollar drop of −$0.9 million. The large difference between the mean and median dollar drop is due to the numerous sizeable outliers occurring in the data. In the period following the announcement, one observes a statistically significant increase in mean CARs. In particular, using the statistical results and supported by the graphical portrayal in Figure 1, Panel A, one

TABLE 5 All breaches (conditioned on announcements in media and not in media) – CAPM-based CARs

This table reports the cumulative abnormal returns for all breaches in the sample determined using the CAPM and conditioned on whether the announcements were reported in the media or were not reported in the media. The event windows include those structured around the announcement date and those in the post-announcement period. The first column indicates the event windows investigated. Four statistical tests were run to assess the significance of the CAPM-based CARs. The second column includes results of the Patell test, the third column describes the results of the standardized cross-sectional test, the fourth column describes the results of the skewness corrected test, and the fifth column includes results of the non-parametric sign test. ** and * denote significance levels of 1% and 5%, respectively.

	In media				Not in media			
Event window	**Patell test**	**Standardized cross-sectional test**	**Skewness corrected test**	**Sign test**	**Patell test**	**Standardized cross-sectional test**	**Skewness corrected test**	**Sign test**
(−3, 1)	−0.86	−0.79	−0.55	−1.84	−1.82	−1.68	−1.50	−0.43
(−2, 1)	−1.80	−1.54	−1.24	−2.49*	−2.38*	−2.04*	−1.91	−1.43
(−2, 0)	−1.65	−1.44	−1.64	−1.41	−1.68	−1.67	−1.29	−0.81
(−2, −1)	−2.26*	−2.07*	−2.02*	−1.84	−1.14	−1.21	−0.90	−0.93
(−1, 1)	−1.09	−0.98	−0.39	−1.19	−2.67**	−2.17*	−2.02*	−0.68
(0)	0.34	0.33	−0.29	0.33	−1.30	−1.29	−1.05	−1.55
(1, 10)	1.14	1.23	1.57	0.76	0.10	0.09	−1.05	0.43
(1, 20)	2.40*	2.40*	2.33*	0.98	0.25	0.25	−0.17	0.56
(2, 10)	1.45	1.56	1.53	1.41	0.72	0.75	−0.42	0.56
(2, 20)	2.63**	2.57*	2.29*	0.98	0.68	0.71	0.35	0.93



Day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mean - Media	0.50	0.44	0.26	0.63	0.74	0.79	0.78	1.07	0.66	0.44	0.38	0.52	0.97	0.94	1.00	0.75	1.25	1.34	1.26	1.21	1.49	1.76	1.91	2.16	2.29	2.16	2.18	2.60	2.43	2.25	2.68
Median - Media	0.24	0.24	0.08	0.13	0.43	0.53	0.48	0.61	0.41	0.30	0.48	0.08	0.17	0.05	0.19	0.36	-0.13	0.02	-0.11	-0.22	0.31	0.86	0.80	1.28	1.93	2.06	1.84	1.98	1.77	1.89	2.17
Mean - WoMedia	0.29	0.49	0.60	0.55	0.39	0.49	0.52	0.60	0.58	0.47	0.37	0.15	0.20	0.18	0.21	0.12	0.13	0.03	-0.05	0.10	0.03	0.13	0.02	0.11	0.20	0.43	0.35	0.22	0.49	0.31	0.29
Median - WoMedia	0.05	0.15	0.23	0.32	0.20	0.26	0.24	0.32	0.24	0.24	0.06	0.08	-0.01	-0.05	-0.10	-0.04	0.36	0.12	0.07	0.58	0.36	0.59	0.51	0.46	0.61	0.67	0.59	0.01	0.38	0.50	0.30

FIGURE 2 All breaches – CAPM CARs. This figure describes the evolution of both the mean and median cumulative abnormal returns for all breaches in the sample calculated using the CAPM from 10 days prior to the announcement of the breach through 20 days following the breach. It separately describes the mean and median of the CARs for the breaches announced in the media and announced without the media. The total number of breaches announced in the media is 85 and those announced without the media are 259, for a total of 344

can observe a small increase of 0.64% (10% significance level) in the mean CARs from day 2 through day 20 (median increase is 0.94%). This corresponds to a mean increase of $613.3 million and a median increase of $74.5 million. Again, outliers cause the large discrepancy between the mean and median. Moreover, this inverse relationship between percentages and dollars is driven by the market cap of the firms that are breached. For example, a large percentage breach can be associated with a relatively small market cap firm. The results using the FF-based mean CARs and presented statistically in Table 4, Panel B, and graphically in Figure 1, Panel B, show strikingly similar results to the CAPM-based results.

Moreover, the results for each of the industry categories analyzed are both statistically and qualitatively similar using both the CAPM-based results and the FF-based results. Hence, only the CAPM-based results are presented in the tables and figures associated with the industry-based results described later in the paper.

There is a significant body of literature demonstrating statistically significant ARs to a target company's stock prior to a merger announcement. While prior knowledge about an upcoming acquisition often results in purchases of the shares of the target by the owners of the non-public information and resulting in upward pressure on the stock price, in the case of a breach, prior knowledge is likely to result in downward pressure on the stock price. Given that the decision of whether and when to report the breach to the media is determined by senior executives, logically one might consider the source of the leak to be related to insiders with knowledge of the breach prior to its announcement as well as knowledge of the expected timing of the announcement. This may well include senior executives and senior cybersecurity personnel who are made aware of the breach. The difference associated with information leakage between

acquisitions and breach announcements revolves around the significantly larger number of parties working on an upcoming acquisition compared to those involved with a breach. An acquisition involves lawyers, investment bankers, accountants, information systems professionals, operations analysts, etc., while the number of parties involved in handling a breach is significantly more limited. Although the graphs in Figure 1 (Panels A and B) present the results for the entire sample of public company breaches, as will be shown in this paper, the same leakage phenomena can be observed when the sample is segmented by breach type and industry. It is interesting to note that such leakage is particularly evident in certain industries and types of breaches that are announced in the media. When the firm reveals the breach in the media, the executives are cognizant of the particular day in which the information will become public (i.e., the announcement day). On the other hand, when the information is passed to the state (e.g., through the state's attorney general), the announcement is not made public through the media, and there is no requirement that the state or its representatives make the information immediately available to the public. Thus, there may well be a lag of several days or longer until the information reaches the public, with the result being that in the case of an announcement of the breach not made through the media, the firm's executives are likely unaware of the specific date that the breach information will be made available to the public. On the other hand, those acting on either direct or indirect knowledge of when a breach announcement will occur in the media may well be the proximate cause of the leakage occurring prior to the announcement.

A further interesting characteristic of the pattern of mean CARs is the increase in the mean CARs in the period immediately following the breach announcement. While initially, it seems counterintuitive that the mean CARs increase following the announcement, a plausible explanation relates to the reassurance generally provided by each of the breached companies following a breach. This increase is generally also exhibited in the various subsamples when the companies are segmented by industry and breach type. This holds in particular for breaches announced in the media. Such results may occur due to the reassurance given by the firm as opposed to a much less publicized filing through state authorities or a centralized database. In other words, in the case of breach announcements in the media, the public is more likely to be exposed to the reassurance that the individual's private information will in the future be protected by enhanced security.

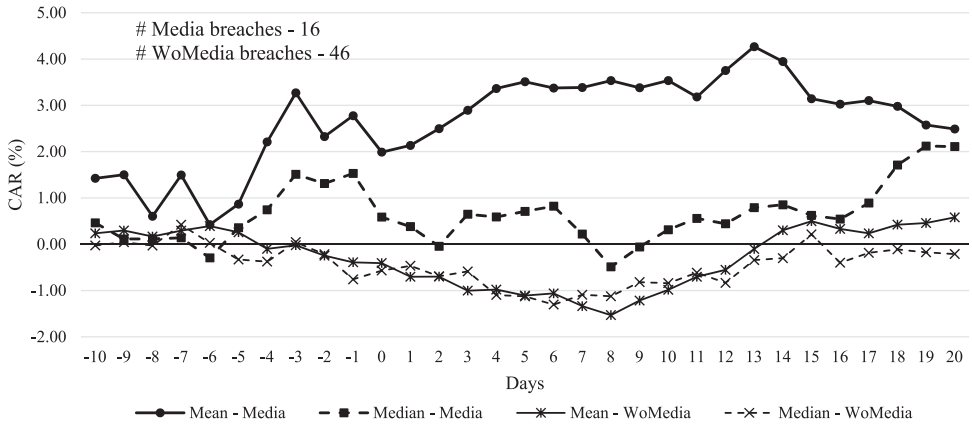
7.2 | Breaches in the media versus breaches out of the media

Similar to the analysis of all breaches, an analysis conditioned on whether the announcement of the breach was in the media (85 breaches) or made through other methods of notification (259 breaches) was also performed. Table 5 presents the statistical results for breaches announced both in the media and those not announced in the media using the CAPM-based CARs. The results suggest that those breaches announced in the media had similar characteristics to the breaches not announced in the media. In other words, both sets of breaches had statistically significant drops in value for 3- and 4-day windows near the announcement date using multiple parametric statistical tests. In addition, for announcements made in the media, drops over the 2-day window (−2, −1) were also statistically significant. Furthermore, the results for the non-parametric sign test were significant for several of the windows surrounding the announcement date of breaches announced in the media, yet none were significant for the breaches not announced in the media. In the longer windows following announcements in the media

TABLE 6 Bank breaches (conditioned on announcements in media and not in media) – CAPM-based CARs

This table reports the cumulative abnormal returns for banks in the sample determined using the CAPM conditioned on whether the announcements were reported in the media or were not reported in the media. The event windows include those structured around the announcement date and those in the post-announcement period. The first column indicates the event windows investigated. Four statistical tests were run to assess the significance of the CAPM-based CARs. The second column includes results of the Patell test, the third column describes the results of the standardized cross-sectional test, the fourth column describes the results of the skewness corrected test, and the fifth column includes results of the non-parametric sign test. ** and * denote significance levels of 1% and 5%, respectively.

	In media				Not in media			
Event window	**Patell test**	**Standardized cross-sectional test**	**Skewness corrected test**	**Sign test**	**Patell test**	**Standardized cross-sectional test**	**Skewness corrected test**	**Sign test**
(−3, 1)	−0.45	−0.45	−0.06	−1.00	−1.34	−1.59	−1.74	−0.29
(−2, 1)	−1.42	−1.03	−1.17	−1.50	−1.87	−1.98*	−1.85	−0.59
(−2, 0)	−1.72	−1.31	−1.66	−1.00	−1.62	−1.94	−1.32	−1.47
(−2, −1)	−0.64	−0.49	−0.80	1.00	−1.24	−1.44	−1.55	−2.36*
(−1, 1)	−0.86	−0.97	−0.32	−1.00	−1.63	−1.96	−1.46	−0.88
(0)	−2.06*	−2.53*	−2.08*	−2.00*	−1.05	−1.17	−0.07	−0.88
(1, 10)	0.71	0.85	1.18	2.00*	−0.78	−0.99	−1.21	−0.88
(1, 20)	0.00	0.01	0.35	−0.50	0.67	0.74	1.33	1.18
(2, 10)	0.70	0.94	1.36	2.00*	−0.51	−0.65	−0.61	−1.18
(2, 20)	−0.03	−0.02	0.32	−0.50	0.90	1.00	1.60	1.18



Day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mean - Media	1.42	1.50	0.60	1.49	0.42	0.87	2.21	3.27	2.33	2.78	1.99	2.13	2.50	2.89	3.37	3.51	3.37	3.38	3.54	3.38	3.54	3.18	3.75	4.27	3.95	3.14	3.02	3.10	2.98	2.58	2.49
Median - Media	0.46	0.11	0.12	0.14	-0.29	0.36	0.74	1.51	1.31	1.53	0.59	0.38	-0.04	0.65	0.59	0.71	0.82	0.22	-0.49	-0.06	0.31	0.56	0.44	0.79	0.85	0.62	0.54	0.89	1.71	2.12	2.11
Mean - WoMedia	0.24	0.30	0.17	0.29	0.39	0.26	-0.10	-0.02	-0.25	-0.39	-0.41	-0.70	-0.70	-1.00	-0.98	-1.11	-1.06	-1.34	-1.53	-1.22	-0.99	-0.70	-0.55	-0.10	0.30	0.50	0.33	0.24	0.42	0.46	0.58
Median - WoMedia	-0.03	0.04	-0.03	0.42	0.03	-0.33	-0.38	0.05	-0.22	-0.76	-0.57	-0.47	-0.69	-0.59	-1.10	-1.13	-1.30	-1.09	-1.12	-0.82	-0.84	-0.61	-0.83	-0.34	-0.30	0.21	-0.40	-0.19	-0.11	-0.18	-0.21

FIGURE 3 Bank breaches – CAPM CARs. This figure describes the evolution of both the mean and median cumulative abnormal returns of bank breaches calculated using the CAPM from 10 days prior to the announcement of the breach through 20 days following the breach. It separately describes the mean and median of the CARs for the breaches announced in the media and announced without the media. The total number of breaches announced in the media is 16 and those announced without the media are 46, for a total of 62

(e.g., (1, 20) and (2, 20)), the CAPM-based mean CARs increase more than the initial decline around the announcement day and become statistically significant.

For breaches not announced in the media, the mean CARs follow the same pattern of negative changes around the announcement and then positive changes for the windows following the breach (e.g., (1, 20)). Yet, these longer window results where the breach announcements were not made in the media are statistically insignificant.

Using the CAPM, two major differences appear in the comparison between breach announcements made in the media and those announced elsewhere. First, note that for a window just prior to the announcement date (−2, −1), the mean CARs for the breaches announced in the media are statistically significant under each of the statistical tests, while for breaches not announced in the media, none of the breaches over that 2-day window have demonstrated statistical significance. This is relevant in that this window is pure leakage. Second, the CAPM-based mean CARs for announcements made in the media are statistically significant for the extended period following the announcement (2, 20), while in the case of breaches announced elsewhere, the post-announcement positive mean CARs are not statistically significant.

The FF-based results for all breaches conditioned on the type of breach (not tabulated) are similar to the CAPM-based results; namely, for breaches announced in the media, the mean CARs are negative for the periods in proximity to the breach announcement. The results in Table 5 support the contention that breaches published in the media get more exposure from the investing public. Moreover, as described previously, the specific identification of the actual announcement date is more uncertain in the case of announcements not made through the media. Thus, these results reflect the lack of knowledge about the specific date on which the

announcement of the breach becomes public through the state or revealed when the breach information is passed along to an organization such as the PRC.

Following the announcement date for breaches revealed in the media, Figure 2 demonstrates that following the announcement the mean and median-based CARs using the CAPM increase more than sufficiently to compensate for the initial decline in the proximity of the announcement date. In particular, following a breach announcement in the media, the mean CARs increase respectively 2.30% (median is 1.69%) from day 1 through day 20. To understand the dollar impact of the recovery, we converted the percentage increase to a mean dollar increase of $1,705.3 million and a median increase of $173.6 million. The large difference is again representing several breaches having an outsized impact on the mean value. For the breaches not announced in the media, there are generally no statistically significant declines in the mean CARs around the announcement date. For these breaches, there also is a non-statistically significant change in the mean CARs in the period following the announcement to either day 10 or day 20 (e.g., (1, 10) or (1, 20)).

7.3 | Bank breaches

The effect of bank breaches announced in the media (16 breaches) using the CAPM (see Table 6) has the interesting characteristic that the window including only the announcement day (day 0) is the only window with at least a 5% level of statistical significance. Other windows around the announcement date, with the exception of (−2, 0) which is modestly significant (10% level using several parametric tests), have negative mean declines in CARs but are not at all statistically significant. Unlike the data for all breaches (Table 4, Panel A), the window containing several days prior to the breach (−2, −1) is statistically insignificant. Yet, the decline on day 0 is significant at the 5% level using the Patell test, the standardized cross-sectional test, the skewness corrected test, and the sign test. In particular, the mean CARs on day 0 drop a statistically significant 0.79%. This corresponds to a mean drop of −$1,630.3 million and median drop of −$1,401.9 million on day 0 using the CAPM. These large drops in both mean and median likely are driven by the importance and sensitivity of bank data and the investor concern related to the breach of security. In the case of bank breaches, the median and mean drop in CARs are both sizeable due to the relative size and homogeneity in the market cap of the breached firms. Following the announcement day, the mean CARs over the 10-day window (1, 10) are positive but insignificant (with the exception of the sign test), with a mean increase of 1.55%. Extending the window to (1, 20) results in an insignificant mean change in the CARs with alternative statistical tests. The outliers on the downside play a role for this window as the mean value increases 0.50%, while the median also increases, though by 1.52%. The results for bank breaches using the CAPM and not revealed in the media (46 breaches) result in a statistically significant decline in mean CARs at the 5% level for the (−2, 1) window using the parametric standardized cross-sectional test and for the non-parametric sign test for (−2, −1). Several other windows were significant at the 10% level, yet most windows were insignificant. The results of these statistical tests can be observed in Table 6.

The CAPM-based mean and median CARs for banks can be observed in Figure 3. Using the FF analysis, the results are strikingly similar to those obtained using the CAPM-based mean CARs (the FF analysis is not presented due to its similarity with the CAPM-based results).

The results are surprising given that bank information is considered by the public to be particularly privacy sensitive. A priori one would expect bank breaches to have a larger impact for longer periods than breaches in other industries. As evidenced by a comparison of the

TABLE 7 Large chain store breaches (conditioned on announcements in media and not in media) – CAPM-based CARs

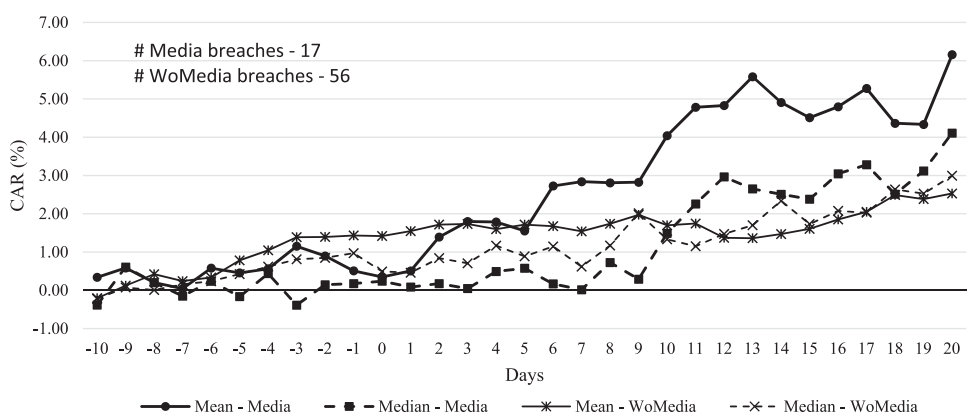
This table reports the cumulative abnormal returns for large chain stores in the sample determined using the CAPM conditioned on whether the announcements were reported in the media or were not reported in the media. The event windows include those structured around the announcement date and those in the post-announcement period. The first column indicates the event windows investigated. Four statistical tests were run to assess the significance of the CAPM-based CARs. The second column includes results of the Patell test, the third column describes the results of the standardized cross-sectional test, the fourth column describes the results of the skewness corrected test, and the fifth column includes results of the non-parametric sign test. ** and * denote significance levels of 1% and 5%, respectively.

	In media				Not in media			
Event window	Patell test	Standardized cross-sectional test	Skewness corrected test	Sign test	Patell test	Standardized cross-sectional test	Skewness corrected test	Sign test
(−3, 1)	−1.14	−1.14	−0.03	−2.18*	0.95	1.05	1.29	0.53
(−2, 1)	−1.73	−1.67	−0.48	−2.67**	0.40	0.43	0.44	−0.80
(−2, 0)	−1.33	−1.20	−0.58	−1.70	−0.01	−0.01	0.11	0.27
(−2, −1)	−1.43	−1.15	−0.62	−2.18*	−0.42	−0.49	0.18	0.53
(−1, 1)	−1.65	−1.50	−0.38	−2.18*	0.75	0.75	0.50	0.00
(0)	−0.28	−0.28	−0.36	−0.24	0.56	0.57	−0.07	−1.34
(1, 10)	2.47*	1.90	1.34	1.21	0.78	0.95	0.58	0.27
(1, 20)	2.30*	1.88	1.58	2.18*	1.38	1.55	1.53	1.34
(2, 10)	2.98**	2.35*	1.43	1.70	0.55	0.64	0.31	−1.34
(2, 20)	0.85	2.20*	1.68	2.18*	−0.11	1.29	1.31	1.07

declines in mean CAPM-based CARs associated with breaches announced in the media for banks (shown in Figure 3) to data for all breaches (Figure 2), one observes that the actual 1-, 2-, and 3-day declines in proximity to the announcement date are similar in magnitude. However, for bank breaches announced in the media using the CAPM, the announcement date is the only window for which the mean CARs are negative and statistically significant at the 5% level. While for all breaches announced in the media, there is leakage at the 5% significance level for the window (−2, −1). Several other windows are significant to a lesser extent. It is interesting that the announcement in the media of a breach in a privacy-sensitive institution such as a bank is not associated with greater magnitude of the breach effects or more statistically significant CARs in any of the windows.

7.4 | Large chain store breaches

The effect of large chain store breaches on mean CAPM-based CARs can be observed in Table 7. First, consider the breach announcements made in the media (17 breaches). For virtually all of the event windows around the announcement date, the CARs are negative, yet generally not statistically significant with the exception of the (−2, 1) window at the 10% level using the parametric Patell test and the standardized cross-sectional test and the 1% level for the non-parametric sign test. For most of the other changes in mean CARs in proximity of the announcement date, the parametric test results are not significant, yet the non-parametric sign test results are significant at the 5% level for the (−1, 1), (−3, 1), and (−2, −1) windows and at the 1% level for the (−2, 1) window. Similar to a number of the other industry classifications,



Day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mean - Media	0.34	0.58	0.20	0.04	0.58	0.46	0.56	1.15	0.90	0.51	0.35	0.51	1.39	1.80	1.78	1.55	2.73	2.84	2.81	2.82	4.04	4.78	4.83	5.58	4.91	4.51	4.80	5.27	4.36	4.33	6.16
Median - Media	-0.39	0.61	0.20	-0.15	0.24	-0.16	0.44	-0.39	0.14	0.18	0.24	0.08	0.17	0.05	0.49	0.58	0.17	0.02	0.73	0.29	1.49	2.26	2.96	2.65	2.51	2.38	3.05	3.28	2.51	3.12	4.11
Mean - WoMedia	-0.21	0.12	0.42	0.24	0.34	0.78	1.04	1.39	1.39	1.43	1.42	1.55	1.72	1.73	1.60	1.72	1.68	1.54	1.74	1.97	1.70	1.74	1.37	1.36	1.47	1.61	1.85	2.05	2.49	2.38	2.53
Median - WoMedia	-0.22	0.07	0.01	0.15	0.24	0.42	0.62	0.81	0.85	0.97	0.49	0.46	0.84	0.71	1.17	0.89	1.15	0.62	1.17	2.01	1.33	1.15	1.47	1.70	2.34	1.73	2.08	2.03	2.64	2.53	3.00

FIGURE 4 Large chain store breaches – CAPM CARs. This figure describes the evolution of both the mean and median cumulative abnormal returns of large chain store breaches calculated using the CAPM from 10 days prior to the announcement of the breach through 20 days following the breach. It separately describes the mean and median of the CARs for the breaches announced in the media and announced without the media. The total number of breaches announced in the media is 17 and those announced without the media are 56, for a total of 73

the post-announcement 10- and 20-day windows are positive and statistically significant using most of the statistical tests.

The results for large chain store breaches not announced in the media indicate that none of the mean CARs is statistically significant. Recall that for the entire sample of breaches not announced in the media, the pattern of negative mean CARs in windows around the breach announcement dates and positive mean CARs in windows following the announcement dates is generally observed. However, it does not hold for the sample of large chain store breaches. For such breach announcements, the mean CARs are generally positive both before and after the announcement date with none being statistically significant.

The graphical depiction of the CAPM-based mean and median CARs are presented in respectively in Figure 4.

Comparing large chain stores to banks for announcements in the media, two observations following the announcements are interesting to note. First, the mean CARs following the announcement of large chain store breaches are about twice as large as bank breaches (6% vs. 3%). One might ask why the data suggests that investors are more reassured following announcements of a breach by a large chain store compared to that of a bank? One hypothesis is that breaches in banks may well have a more detrimental potential downside if made public (e.g., account numbers etc.) than breaches in large chain stores. Given the historically large number of multiple bank breaches (see Table 8), investors may believe that banks are prone to revealing more serious information through these multiple breaches than large chain stores, and as a result, have less credibility that their personal information will be secure.

The second observation is that following the announcements made in the media, banks have virtually no windows that exhibit statistically significant mean CARs as opposed to large chain store breaches with the mean CARs for nearly all windows being statistically significant. Given the lack of statistically significant positive change in mean CARs following breach announcements, the reassurance affecting other industries is not being accepted by bank investors. This contrasts dramatically with large chain stores where the reassurance effect is

TABLE 8 Firms with more than five breaches in the sample

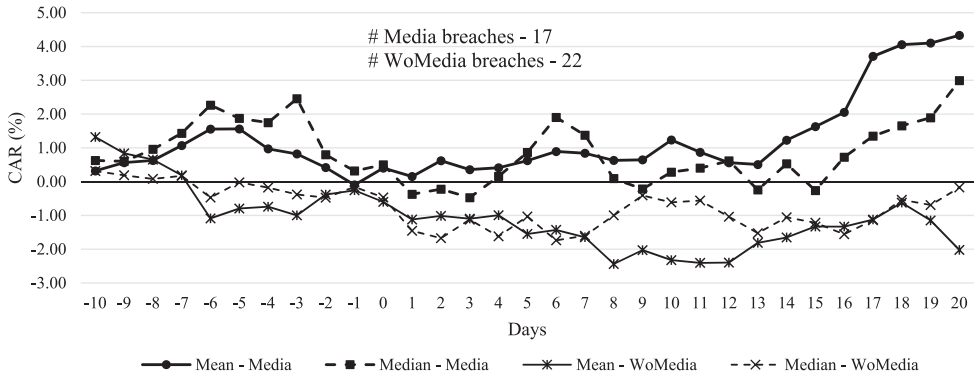
This table reports firms with more than five breaches in the sample. Column 1 indicates the firms in respective banking and large chain stores with more than five breaches. Column 2 indicates the number of breaches associated with each firm.

Company	# Breaches
Banks with >5 breaches	
Bank of America	12
JPMorgan Chase	12
Citigroup	8
Wells Fargo	6
Large chain stores with >5 breaches	
CVS	10
Walgreens	9
McDonald's	9
Home Depot	7
Sears	7

TABLE 9 Technology breaches (conditioned on announcements in media and not in media) – CAPM-based CARs

This table reports the cumulative abnormal returns for technology firms in the sample determined using the CAPM conditioned on whether the announcements were reported in the media or were not reported in the media. The event windows include those structured around the announcement date and those in the post-announcement period. The first column indicates the event windows investigated. Four statistical tests were run to assess the significance of the CAPM-based CARs. The second column includes results of the Patell test, the third column describes the results of the standardized cross-sectional test, the fourth column describes the results of the skewness corrected test, and the fifth column includes results of the non-parametric sign test. ** and * denote significance levels of 1% and 5%, respectively.

	In media				Not in media			
Event window	**Patell test**	**Standardized cross-sectional test**	**Skewness corrected test**	**Sign test**	**Patell test**	**Standardized cross-sectional test**	**Skewness corrected test**	**Sign test**
(−3, 1)	−0.96	−0.91	−0.77	−0.73	−1.29	−1.36	−0.29	−1.71
(−2, 1)	−1.17	−1.25	−0.67	−0.73	−1.09	−1.18	−0.09	−1.28
(−2, 0)	−0.79	−0.90	−0.57	−0.24	−0.34	−0.37	0.41	−1.28
(−2, −1)	−1.65	−2.11*	−1.61	−1.21	0.30	0.32	1.01	−0.43
(−1, 1)	−0.85	−0.97	−0.29	−0.73	−1.82	−1.83	−0.77	−1.28
(0)	0.97	0.94	0.98	0.24	−1.01	−1.24	−0.56	−0.85
(1, 10)	0.14	0.16	0.63	−0.24	−0.48	−0.54	−1.02	0.43
(1, 20)	1.89	1.59	1.77	0.24	−0.39	−0.49	−0.66	0.43
(2, 10)	0.47	0.54	0.98	0.73	0.03	0.04	−0.82	−0.43
(2, 20)	2.16*	1.78	1.87	0.73	−0.03	−0.03	−0.46	0.43



Day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mean - Media	0.32	0.56	0.63	1.07	1.56	1.56	0.97	0.82	0.42	-0.10	0.40	0.15	0.62	0.36	0.41	0.62	0.89	0.84	0.63	0.65	1.23	0.87	0.56	0.51	1.23	1.63	2.05	3.71	4.06	4.10	4.33
Median - Media	0.63	0.60	0.96	1.43	2.27	1.87	1.75	2.46	0.80	0.32	0.50	-0.37	-0.22	-0.47	0.16	0.86	1.90	1.37	0.09	-0.22	0.28	0.40	0.62	-0.24	0.53	-0.26	0.72	1.35	1.65	1.89	2.99
Mean - WoMedia	1.32	0.84	0.65	0.19	-1.09	-0.80	-0.74	-1.00	-0.38	-0.26	-0.60	-1.12	-1.01	-1.10	-0.99	-1.55	-1.43	-1.65	-2.44	-2.03	-2.32	-2.40	-2.40	-1.81	-1.65	-1.33	-1.33	-1.12	-0.62	-1.15	-2.02
Median - WoMedia	0.31	0.19	0.08	0.17	-0.47	-0.02	-0.18	-0.38	-0.47	-0.16	-0.47	-1.46	-1.67	-1.12	-1.62	-1.03	-1.74	-1.60	-1.00	-0.41	-0.61	-0.56	-1.04	-1.52	-1.05	-1.22	-1.56	-1.14	-0.54	-0.69	-0.17

FIGURE 5 Technology breaches – CAPM CARs. This figure describes the evolution of both the mean and median cumulative abnormal returns of technology breaches calculated using the CAPM from 10 days prior to the announcement of the breach through 20 days following the breach. It separately describes the mean and median of the CARs for the breaches announced in the media and announced without the media. The total number of breaches announced in the media is 17 and those announced without the media are 22, for a total of 39

significant and post-announcement the mean CARs increase dramatically and statistically significantly. For those breaches not announced in the media, none of the windows for large chain stores are statistically significant, and virtually no bank windows following the announcement are statistically significant.

7.5 | Technology company breaches

The effect of technology breaches announced in the media (17 breaches) on CAPM-based mean CARs can be seen in Table 9. These results are similar to those obtained for the entire sample of breaches. For the CAPM-based mean CARs, the only statistically significant results in the sample of announcements made in the media occurred for the (−2, −1) window. Like the sample of all breaches announced in the media, it is interesting to note that the only window in proximity to the announcement date with statistically significant results at the 5% level occurs for the (−2, −1) window. As with a number of other industries, the (−2, −1) window consists entirely of a period of leakage, that is, prior to the announcement. For example, the drop in the mean CARs in the period (−2, −1) is a statistically significant 0.92% (median is 2.14%) using the standardized cross-sectional test. This corresponds to a mean and median drop of respectively −$1,670 million and−$653 million using the CAPM.[1] In the period following the announcement date, the results are

[1]It is not uncommon for the larger median percentage change to be associated with a smaller dollar median change. This can result from the fact that a relatively large percentage change in a security can be associated with a small cap security and as a result have a negligible dollar effect. The converse effect also may hold, where a small percentage change can have a large dollar effect.

TABLE 10 Hacking (conditioned on announcements in media and not in media) – CAPM-based CARs

This table reports the cumulative abnormal returns for hacked firms in the sample determined using the CAPM conditioned on whether the announcements were reported in the media or were not reported in the media. The event windows include those structured around the announcement date and those in the post-announcement period. The first column indicates the event windows investigated. Four statistical tests were run to assess the significance of the CAPM-based CARs. The second column includes results of the Patell test, the third column describes the results of the standardized cross-sectional test, the fourth column describes the results of the skewness corrected test, and the fifth column includes results of the non-parametric sign test. ** and * denote significance levels of 1% and 5%, respectively.

	In media				Not in media			
Event window	**Patell test**	**Standardized cross-sectional test**	**Skewness corrected test**	**Sign test**	**Patell test**	**Standardized cross-sectional test**	**Skewness corrected test**	**Sign test**
(−3, 1)	−1.36	−1.12	−0.74	−1.26	−2.57*	−2.22*	−2.06*	−1.90
(−2, 1)	−2.01*	−1.68	−1.24	−1.90	−2.06*	−1.54	−1.56	−1.31
(−2, 0)	−2.09*	−1.75	−1.86	−1.58	−1.53	−1.32	−1.33	−1.31
(−2, −1)	−1.90	−1.74	−1.64	−1.26	0.15	0.17	−0.06	−0.15
(−1, 1)	−1.67	−1.35	−0.66	−0.95	−2.82**	−1.96*	−1.79	−1.60
(0)	−0.93	−0.81	−1.31	−0.63	−2.86**	−2.15*	−2.12*	−2.48*
(1, 10)	0.90	0.83	1.49	0.63	1.03	0.99	0.08	0.73
(1, 20)	2.68**	2.23*	2.02*	0.95	0.37	0.38	−0.33	0.44
(2, 10)	1.09	1.02	1.29	0.63	1.58	1.48	0.43	1.02
(2, 20)	2.85**	2.32*	1.88	1.26	0.72	0.73	−0.04	1.02

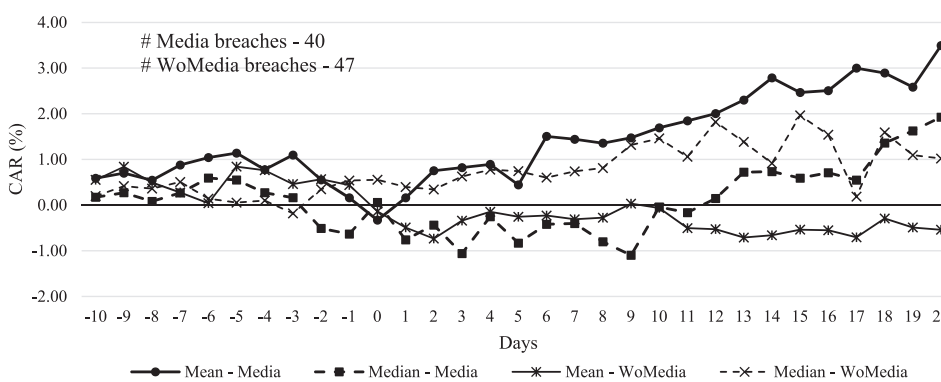
statistically significant at the 10% level for (1, 20) for several of the statistical tests, but at the 5% level for the (2, 20) window using the Patell test and at the 10% level using the standardized cross-sectional test and skewness corrected test. Thus, overall, once again, the decline in the mean CARs during the period (−2, −1) suggests that leakage is prevalent in technology breaches when the breach is announced in the media. Moreover, the market accepts the reassurance provided at the time of the announcement. These results are reinforced by the fact that virtually none of the mean CARs are statistically significant for any of the windows when the breach announcements are not made in the media (22).

Although the results were generally not statistically significant for announcements not made in the media, the pattern of results was similar to most other industries in the period in proximity to the announcement date, namely a drop in CAPM-based mean CARs. Yet, in the 10–20 day period following the announcement, the CARs were frequently negative, although statistically insignificant.

Figure 5 is a graphical representation of the mean and median CARs generated by the CAPM.

7.6 | Other industry category breaches

Several other industry categories were analyzed, namely telecommunications breaches and other financial institution breaches. For both of these categories, the number of breach announcements occurring in the media was insufficient to perform reliable statistical tests. In the telecommunications industry and other financial institutions, there were respectively 6 and 12 breaches announced in the media, which were both deemed to be insufficient (see Table 3). As



Day	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mean - Media	0.59	0.70	0.54	0.88	1.04	1.14	0.78	1.10	0.55	0.16	-0.33	0.16	0.75	0.82	0.89	0.44	1.51	1.44	1.36	1.47	1.69	1.85	2.00	2.30	2.79	2.47	2.51	3.00	2.89	2.58	3.49
Median - Media	0.17	0.27	0.08	0.27	0.59	0.55	0.27	0.16	-0.51	-0.63	0.06	-0.76	-0.44	-1.06	-0.25	-0.83	-0.42	-0.40	-0.80	-1.10	-0.04	-0.16	0.15	0.72	0.73	0.59	0.71	0.55	1.36	1.62	1.93
Mean - WoMedia	0.55	0.84	0.49	0.28	0.04	0.84	0.76	0.46	0.57	0.44	-0.14	-0.49	-0.73	-0.34	-0.15	-0.26	-0.23	-0.31	-0.28	0.03	-0.06	-0.50	-0.53	-0.71	-0.66	-0.54	-0.55	-0.71	-0.29	-0.49	-0.54
Median - WoMedia	0.20	0.42	0.36	0.51	0.14	0.05	0.10	-0.18	0.35	0.54	0.55	0.40	0.34	0.63	0.78	0.74	0.60	0.74	0.81	1.32	1.46	1.06	1.82	1.38	0.92	1.96	1.54	0.18	1.59	1.09	1.02

FIGURE 6 Hacking breaches – CAPM CARs. This figure describes the evolution of both the mean and median cumulative abnormal returns of hacking breaches calculated using the CAPM from 10 days prior to the announcement of the breach through 20 days following the breach. It separately describes the mean and median of the CARs for the breaches announced in the media and announced without the media. The total number of breaches announced in the media is 40 and those announced without the media are 47, for a total of 87

discussed earlier, the minimum required to statistically analyze the breach category was 15 breaches.[2] While breaches in the telecommunications and other financial institutions industries that were not announced in the media were sufficient to analyze, the results were similar enough to several other categories, and therefore we describe them here without presenting the statistical and graphical analysis. First, consider the telecommunications breaches (18 breaches). The breaches around the announcement date using the CAPM resulted in drops in the mean CARs that were statistically significant for a number of windows (e.g., (−2, 1), (−2, 0), (−2, −1), (−1, 1)), yet the only drops that were significant at better than a 10% significance level were (−3, 1) and (−2, 0). These were at the 5% significance level as measured by respectively, the sign test and the standardized cross-sectional test. Following the announcement date, the 10-day window of mean CARs (e.g., (1, 10)) was significantly positive at the 1% level using the standardized cross-sectional test and the (2, 10) window was also significant at the 1% level using other statistical tests, while the 20-day mean CARs (e.g., (1, 20)) were not at all statistically significant. The FF-based mean CARs produced virtually identical results.

Around the announcement date, the pattern of CARs resulting from breaches at other financial institutions whose announcements were not in the media is very similar to those for technology firm breaches not announced in the media. There generally is a non-statistically significant drop in proximity to the announcement date. Following the announcement and associated drop in mean CARs, the patterns diverge. For telecommunications firms, the mean CARs begin to increase, yet with several exceptions, they are generally not significant. For other financial institutions, the mean CARs generally fall following the announcement, yet are each not significant. The results are similar for both the CAPM and FF-based analyses.

7.7 | Breach types: Hacking

The effect of hacking on CAPM-based CARs is demonstrated in Table 10 (40 hacks). Like many of the industry-based analyses, the CARs for windows near the announcement date for announcements made in the media are generally negative, with some windows significant at either the 5% level or 10% level for certain statistical tests. For example, the mean CARs for windows (−2, 1) and (−2, 0) are both significant at the 5% level using the Patell test. However, using the other tests they are either significant at 10% or not significant at all. Moreover, the drop over the (−2, 0) window is 1.43%, with a median CAR drop of 0.10%. The mean dollar drop corresponds to −$478.6 million with the media dropping −$59.3 million. The window including leakage during days −2 and −1 (namely (−2, −1)) is significant at the 10% level using the Patell test and the standardized cross-sectional test, with a mean drop of 0.94% (median drop is 0.79%). This corresponds to a mean dollar drop of −$653.9 million and a median drop of −$14.9 million. The post-announcement 20-day windows (e.g., (1, 20) and (2, 20)) shown in the table are generally significant at either the 1% level or the 5% level, with the increase in mean CARs for the (1, 20) window being 3.99% (median is 2.12%) corresponding to a mean dollar increase of $3,440.0 million and a median increase of $146.8 million. The large spread between the mean and median is driven by a number of sizeable hacking outliers.

The pattern of the CAPM-based mean declines in CARs for hacking breaches not announced in the media is similar to those announced in the media for most of the preannouncement windows. The mean CARs for most of the preannouncement windows observed are negative, and often statistically significant for those hacks not announced in the media. Several partially overlapping

[2]Since the miscellaneous category was not categorized by industry and we had analyzed the combined breach category, a separate analysis of the miscellaneous category was not performed.

windows are also associated with negative and statistically significant windows for those breaches announced in the media. These can be observed in Table 10. Yet, it is particularly noteworthy that for hacking announcements made in the media, leakage is evident in the mean CAPM-based decline in CARs for the (−2, −1) window. This is not observed for announcements not made in the media. Following the announcement, the mean CARs over the 20-day window for announcements made in the media are often positive and statistically significant, while those not made in the media are sometimes negative and not statistically significant. The mean and median CARs can be observed in Figure 6. The dominance of the negative outliers drives variability and causes the lack of statistical significance for the category of hacks not announced in the media.

Using FF-based mean CARs results in virtually identical statistical results and graphical analysis as the CAPM-based mean CARs. This results for both breaches announced in the media and breaches not announced in the media.

7.8 | Other breach types: Phishing, theft, and negligence

Unlike the case of breaches caused by hacking, in general, phishing, theft, and negligence breaches have little or no statistically significant impact on the mean CARs. For phishing, the only breaches included in our statistical analysis are those not announced in the media. The reason is that out of the 10 phishing windows analyzed, only announcements not made in the media had sufficient numbers to analyze. No windows in the proximity of the announcement date were statistically significant, with some windows having positive changes in mean CARs and others having negative changes. Post-announcement, the mean CARs were all positive, with the windows (1, 20) and (2, 20) being statistically significant at the 5% level.

In the case of theft, considering both announcements made in the media and not made in the media, the only window having any statistical significance (though only at the 10% level) is (−2, −1) in the case of announcements made in the media. For the case of negligence, for both breaches announced in the media and not in the media, only one was statistically significant, but at the 10% level of significance.

8 | CONCLUSIONS

While computer breaches have received significant attention in the financial press and mainstream media, little analysis is available in the academic literature. Our analysis of the market effect of breaches has provided insight into the effects of breach type and the industry of the breached company. Overall, the results are surprising. In general, using an event study methodology and determining CARs, in the period prior to the breach, the value of the breached company often significantly declines. Counterintuitively, in general, in the period following the breach announcement, the CARs are typically positive and increasing. Moreover, when considering all breach types together, the CARs for breaches announced in the media are statistically significant in periods both before and after the announcement date. This is in contrast to breaches not announced in the media, where the CARs are only significant in the period before the announcement date. Interestingly, when comparing the results for those breaches announced in the media and those not announced in the media, one can observe pure leakage (a window prior to the announcement date) in the former, and no leakage in the latter. In other words, when company management is aware of the likely date of the announcement, there is statistically significant pure leakage, while in the situation where management is not cognizant of the exact date of the announcement, there is no such leakage.

The pattern of negative and decreasing CARs before the announcement and positive and increasing CARs following the announcement is also observed when segmenting the sample by industry. However, one surprising result is that the CARs following the announcement for large chain stores are positive at approximately 6% as opposed to 3% for banks. Though one might have expected technology companies to be more sophisticated in their cybersecurity and thus breaches in this industry to have more impact on their value, except for the window of the announcement day itself, the results prior to the announcement are not significant, with no evident leakage.

When segmenting the breaches by breach type, one observes that the breach type with the most noticeable impact on value is hacking (when the hack is announced in the media), which exhibits a statistically significant decline prior to the announcement, followed by a statistically significant increase in the post-announcement period. In general, with the exception of breaches categorized as phishing, with significant decline in the 2-day window prior to the announcement date, the results for the other types of breaches (theft and negligence) do not exhibit a statistically significant value impact.

The positive CARs in the period following a security breach announcement have important implications for analysts, portfolio managers, and the investing community. If at all, the inclination is to divest such a company's stock, while the results presented in the article suggest an opposite action. Furthermore, the negative ARs in the weeks before a security breach announcement warrant an SEC forensic analysis of leakage-driven trades prior to such announcements.

There are a number of topics associated with security breaches that are appropriate for future research. An interesting related area is the study of firms in which security breaches have occurred multiple times. For example, Bank of America was breached 12 times during our study period. The effect of both multiple breaches and the size of the breaches on CARs would be a useful analysis. Similarly, it would be useful for understanding the timing of the breaches; for example, is there more of an effect from back to back breaches than breaches that are years apart. Moreover, it would also be interesting to understand the sequential effect of breaches on the associated CARs. Namely, does the second breach at a firm have a larger CAR effect than the first, etc. It would be also useful to understand whether there are industry breach effects. For example, does a breach in an industry have any significant effect on other firms in the industry? In other words, does a breach suggest that other firms in the same industry have an increased likelihood of occurring (i.e., is reflected in the other firms' CARs)? Or could it suggest that other firms in the industry might benefit from a competitor's breach? An additional topic for future research might be the co-movement of breached firms and data security firms. As firms are breached, there may well be a positive effect on those firms providing data security.

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